

Egyptian Compression Length — ESC / Sierpiński / Goldbach Diagnostic

Folder: focus_shima **Status:** working note. Definitions are settled; the conjectures are open; the Goldbach link is diagnostic, not a proof bridge.

1. Core object: the unit-fraction length

Stop writing the repeated sum. Define the set of values reachable with exactly $\boxed{t}$ positive unit fractions:

$$\mathcal{E}_t = \left\{ \sum_{i=1}^t \frac{1}{a_i} : a_i \in \mathbb{Z}_{>0} \right\}$$

(no distinctness requirement; repeats allowed). Then define the **Egyptian compression length**:

$$\boxed{L_{\text{UF}}(x) = \min\{t : x \in \mathcal{E}_t\}}$$

Every positive rational has a finite expansion, so L_{UF} is always defined and finite. The interesting object is therefore **not** infinity but the **bounded-length** question $L_{\text{UF}}(x) \leq t$.

For completeness:

$$\mathcal{E}_* = \bigcup_{t \geq 1} \mathcal{E}_t \quad (= \text{all positive rationals, by finiteness of Egyptian expansions})$$

2. The conjectures, restated in one line each

Name	Compact form	Status
Erdős-Straus	$L_{\text{UF}}(4/n) \leq 3$ for all $n \geq 2$	open
Sierpiński	$L_{\text{UF}}(5/n) \leq 3$ for all $n \geq 2$	open
Schinzel (general)	$L_{\text{UF}}(k/n) \leq 3$ for all n large enough (given k)	open
3-slot problem	$L_{\text{UF}}(k/n) \leq 3$	family of hard Diophantine cases

Note the structural point: ****the number of slots is fixed at 3.**** What moves across these conjectures is the numerator k , not the slot count. So $4 = 3 + 1$ is **not** a moving identity $k = t + 1$ — Sierpiński keeps $t = 3$ and takes $k = 5$.

3. Why $k = t + 1$ is the threshold of first difficulty

The intuition " $k = t + 1$ " is correct as the **first hard numerator** for a given slot count t , and this is exactly why $4/n$ (at $t = 3$) is the canonical case.

Fix t slots. Any numerator $k \leq t$ is **trivial**, because the unary default already fits:

$$\frac{k}{n} = \underbrace{\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n}}_{k \text{ copies}} \in \mathcal{E}_k \subseteq \mathcal{E}_t \quad (k \leq t).$$

So for $t = 3$:

- $1/n$, $2/n$, $3/n$ are all trivial. In particular $3/n = 1/n + 1/n + 1/n$ uses exactly the 3 slots with nothing to spare.
- $4/n$ is the **first numerator whose unary default overflows** — it wants 4 slots, only 3 are available. First case requiring genuine recombination, not just copying.

Threshold pattern, and how the verdict depends on t :

- $t = 2$: threshold $k = 3$ is already **false** — e.g. $3/7$ has no two-term unit-fraction representation.
- $t = 3$: threshold $k = 4$ is **open** (conjecturally true) — Erdős–Straus.

So $k = t + 1$ is the first overflow at every t ; what changes is whether it holds.

4. The unary basis / repunit reading (why repetitions keep returning)

Unit fractions are a **unary basis**. The dumb default representation of k/n is literally k repetitions of $1/n$ — the repunit-style "uncompressed code." Erdős–Straus is the claim that the unary block

$$\underbrace{1\,1\,1\,1}_{k=4} \longrightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

can always be compressed into **three reciprocal atoms**. The repetition layer keeps reappearing because the original problem is *built from* unary atoms; you are trying to escape the layer the object is made of.

It does not vanish under inversion — it changes costume. Clearing denominators in $\frac{k}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$ gives

$$kabc = n(ab + ac + bc).$$

The right side is a symmetric pile of pairwise products $ab + ac + bc$ — the hidden repetition/product channel. The unary structure is preserved on the multiplicative side.

****Design principle for any compression we adopt:**** the shorthand must preserve this difficulty. $k/n \in \mathcal{E}_t$ and $L_{\text{UF}}(k/n) \leq t$ both do. A private shorthand like " $k/n = t/(n_*)$ " should be avoided in formal text — it reads as an ordinary fraction rather than " t unit-fraction slots." If a slot operator is wanted, prefer $k/n \Rightarrow_t 1/*$, but $\in \mathcal{E}_t$ is cleaner.

5. The inversion gate (multiplicative obstruction)

Force the first denominator as $a = nt$. Then the remainder is two-term:

$$\frac{4}{n} - \frac{1}{nt} = \frac{4t - 1}{nt} = \frac{1}{b} + \frac{1}{c}.$$

Standard divisor inversion of the two-term remainder gives

$$((4t - 1)b - nt)((4t - 1)c - nt) = n^2t^2.$$

So a solution is equivalent to finding a divisor pair $r, s \mid n^2t^2$ with $rs = n^2t^2$ such that

$$b = \frac{r + nt}{4t - 1}, \quad c = \frac{s + nt}{4t - 1}$$

are both positive integers.

ESC inversion gate:

Layer	Meaning
n	number being tested
t	first-denominator scale, $a = nt$
$4t - 1$	inversion modulus
$r, s \mid n^2t^2$	divisor shell
failure	no divisor pair satisfies integrality / congruence condition

This is a clean **obstruction checker**, multiplicative in character.

6. Why it is genuinely hard — the arithmetic obstruction

The threshold overflow alone would be easy; the difficulty is arithmetic. It suffices to prove ESC for primes. Mordell reduced the remaining cases to the residues

$$n \equiv 1, 121, 169, 289, 361, 529 \pmod{840}, \qquad 840 = 2^3 \cdot 3 \cdot 5 \cdot 7.$$

These six are exactly the **squares** of residues coprime to 840. So the hard fibers carry a **quadratic-residue (square-class) signature** — which is why elementary modular covering systems keep missing them. The obstruction is not a counting overflow; it is a square-class condition mod 840.

This is the load-bearing fact for treating ESC as a diagnostic: the signal it carries is *square-class structure*, orthogonal to additive Hardy-Littlewood residue structure.

7. Goldbach application — prime-fiber Egyptian obstruction gate

Caveat first. At the raw Goldbach level $n = N = p + q$ the gate degenerates, because N is even and even N are trivial:

$$\frac{4}{N} = \frac{2}{m} = \frac{1}{m} + \frac{1}{m+1} + \frac{1}{m(m+1)} \quad (N = 2m).$$

So ESC gives almost no obstruction on the even sum itself; it is already closed by parity. The nontrivial obstruction space lives on **odd primes / residue classes**.

Use it on the prime fibers, not on N . For each Goldbach fiber $p + q = N$, compute an obstruction score $O_{\text{ESC}}(\cdot)$ via the divisor-inversion gate, on p, q , or nearby shifted forms $p \pm 1, q \pm 1, N - p, N - q$. Then compare

$$\sum_{p+q=N} O_{\text{ESC}}(p) O_{\text{ESC}}(q)$$

against the Hardy-Littlewood / Ramanujan baseline.

What this tests: whether reciprocal-divisor (multiplicative, square-class) obstruction adds signal beyond the additive Goldbach residue structure. The honest framing is a **two-channel comparison**, not a proof bridge:

additive Fourier / Ramanujan channel
$$R(N)=\int_0^1 S(\alpha)^2 e(-N\alpha) \, d\alpha$$

vs.

multiplicative Egyptian-fraction inversion channel
$$\text{ESC divisor gate}$$

8. Verdict table

Claim	Verdict
Use ESC at $N = p + q$ directly	weak / trivial — N even, closed by parity
Use ESC on prime fibers (p, q)	good diagnostic
Inversion formulas as obstruction gates	clean, usable
Prove Goldbach from ESC (or vice versa)	no known bridge
Treat as unit-circle closure theorem	too strong — false framing
Generalize to k/n	possible, but a family of hard Diophantine problems

Honest read: promising as a classifier / diagnostic (additive vs. multiplicative obstruction spectra), not as a proof path. The pre-registered expectation should be that the multiplicative channel carries square-class signal that is largely orthogonal to — and may or may not beat — the Hardy-Littlewood baseline. A clean negative (no signal beyond baseline) is a publishable result.

Quick reference

$$\mathcal{E}_t = \left\{ \sum_{i=1}^t 1/a_i : a_i \in \mathbb{Z}_{>0} \right\}, \quad L_{\text{UF}}(x) = \min\{t : x \in \mathcal{E}_t\}$$

ESC: $L_{\text{UF}}(4/n) \leq 3 \ \forall n \geq 2$

Threshold: $k \leq t$ trivial, $k = t + 1$ first hard